
Problem Set 6

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Problem 1

1. This problem is solved by fully decomposing the given state $|\psi_0\rangle$. We have

$$\begin{aligned} |\psi_0\rangle &= |+\rangle_{\theta;1} \otimes |-\rangle_{\theta;2} + A|-\rangle_{\theta;1} \otimes |+\rangle_{\theta;2} \\ &= (\cos\theta|+\rangle_1 + \sin\theta|-\rangle_1) \otimes (-\sin\theta|+\rangle_2 + \cos\theta|-\rangle_2) + A(-\sin\theta|+\rangle_1 + \cos\theta|-\rangle_1) \\ &\quad \otimes (\cos\theta|+\rangle_2 + \sin\theta|-\rangle_2) \\ &= -\cos\theta\sin\theta(|+\rangle_1 \otimes |+\rangle_2) + \cos^2\theta(|+\rangle_1 \otimes |-\rangle_2) - \sin^2\theta(|-\rangle_1 \otimes |+\rangle_2) \\ &\quad + \sin\theta\cos\theta(|-\rangle_1 \otimes |-\rangle_2) - A\sin\theta\cos\theta(|+\rangle_1 \otimes |+\rangle_2) - A\sin^2\theta(|+\rangle_1 \otimes |-\rangle_2) \\ &\quad + A\cos^2\theta(|-\rangle_1 \otimes |+\rangle_2) + A\cos\theta\sin\theta(|-\rangle_1 \otimes |-\rangle_2) \\ &= (-\cos\theta\sin\theta - A\cos\theta\sin\theta)|++\rangle + (\cos^2\theta - A\sin^2\theta)|+-\rangle + (-\sin^2\theta + A\cos^2\theta)|-+\rangle \\ &\quad + (\cos\theta\sin\theta + A\cos\theta\sin\theta)|--\rangle. \end{aligned}$$

Note that by setting $A = -1$, we have

$$|\psi_0\rangle = 0|++\rangle + |+-\rangle - |-+\rangle + 0|--\rangle = |+-\rangle - |-+\rangle.$$

Note that this is independent of θ , so $A = -1$.

2. First, we compute $S_z|\psi_0\rangle$.

$$\begin{aligned} S_z|\psi_0\rangle &= \left(S_z^{(1)} \otimes \mathbb{1} + \mathbb{1} \otimes S_z^{(2)}\right) (|+\rangle_1 \otimes |-\rangle_2 - |-\rangle_1 \otimes |+\rangle_2) \\ &= \frac{\hbar}{2}|+\rangle_1 \otimes |-\rangle_2 + \frac{\hbar}{2}|-\rangle_1 \otimes |+\rangle_2 - \frac{\hbar}{2}|+\rangle_1 \otimes |-\rangle_2 - \frac{\hbar}{2}|-\rangle_1 \otimes |+\rangle_2 \\ &= 0. \end{aligned}$$

Next, we compute $S_x|\psi_0\rangle$.

$$\begin{aligned} S_x|\psi_0\rangle &= \left(S_x^{(1)} \otimes \mathbb{1} + \mathbb{1} \otimes S_x^{(2)}\right) (|+\rangle_1 \otimes |-\rangle_2 - |-\rangle_1 \otimes |+\rangle_2) \\ &= \frac{\hbar}{2}|-\rangle \otimes |-\rangle - \frac{\hbar}{2}|+\rangle_1 \otimes |+\rangle_2 + \frac{\hbar}{2}|+\rangle_1 \otimes |+\rangle_2 - \frac{\hbar}{2}|-\rangle_1 \otimes |-\rangle_2 \\ &= 0. \end{aligned}$$

Finally, we compute $S_y|\psi_0\rangle$. Since we don't often compute things in S_y , note that $S_y|+\rangle = \frac{i\hbar}{2}|-\rangle$, and $S_y|-\rangle = \frac{-i\hbar}{2}|+\rangle$.

$$\begin{aligned} S_y|\psi_0\rangle &= \left(S_y^{(1)} \otimes \mathbb{1} + \mathbb{1} \otimes S_y^{(2)}\right) (|+\rangle_1 \otimes |-\rangle_2 - |-\rangle_1 \otimes |+\rangle_2) \\ &= \frac{i\hbar}{2}|-\rangle_1 \otimes |-\rangle_2 + \frac{i\hbar}{2}|+\rangle_1 \otimes |+\rangle_2 - \frac{i\hbar}{2}|+\rangle_1 \otimes |+\rangle_2 - \frac{i\hbar}{2}|-\rangle_1 \otimes |-\rangle_2 \\ &= 0. \end{aligned}$$

Problem 2

1. The Hamiltonian has two distinct terms, $-2JS_z^{(1)} \otimes S_z^{(2)}$, and $g(S_z^{(1)} + S_z^{(2)})$. By linearity of multiplication/inner product/whatever, we can calculate

$$H_{ij} = \langle b_j | \left(-2JS_z^{(1)} \otimes S_z^{(2)} \right) | b_i \rangle + g \langle b_j | \left(S_x^{(1)} + S_x^{(2)} \right) | b_i \rangle$$

for $|b_i\rangle, |b_j\rangle$ the i th and j th basis states, respectively. I will first do the calculations in the first term, since they are very simple. Let $|a\rangle_1 \otimes |a\rangle_j$ and $|b\rangle_1 \otimes |b\rangle_2$ represent arbitrary basis states. Then

$$\begin{aligned} -2J (\langle a|_1 \otimes \langle a|_2) \left(S_z^{(1)} \otimes S_z^{(2)} \right) (|b\rangle_1 \otimes |b\rangle_2) &= -2J_1 \langle a| S_z^{(1)} |b\rangle_1 {}_2 \langle a| S_z^{(2)} |b\rangle_2 \\ &= -2J b_{1\ 1} \langle a|b\rangle_1 b_{2\ 2} \langle a|b\rangle_2, \end{aligned}$$

where b_i is the eigenvalue of S_z corresponding to $|b\rangle_i$, because each $|a\rangle_i, |b\rangle_i$ is either $|+\rangle$ or $|-\rangle$, which are eigenstates of S_z . Moreover, since these states are orthonormal,

$$-2J b_{1\ 1} \langle a|b\rangle_1 b_{2\ 2} \langle a|b\rangle_2 = -2J b_1 b_2 \delta_{ab},$$

where δ_{ab} is an abuse of notation on the Kroenecker delta to say if $|a\rangle_1 \otimes |a\rangle_2 = |b\rangle_1 \otimes |b\rangle_2$, then $\delta_{ab} = 1$, and 0 otherwise. Since $|+\rangle$ has eigenvalue $\hbar/2$ and $|-\rangle$ has an eigenvalue of $-\hbar/2$ on S_z , this part of the summation can easily be seen to be

$$-\frac{\hbar^2 J}{2} \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & 1 \end{pmatrix}.$$

Now to compute the second term. I will show work for the $S_x^{(1)}$ term only, since the work for the $S_x^{(2)}$ term is basically equivalent. Let $|a\rangle_1 \otimes |a\rangle_2$ and $|b\rangle_1 \otimes |b\rangle_2$ be basis vectors, and note that

$$(\langle a|_1 \otimes \langle a|_2) S_x^{(1)} (|b\rangle_1 \otimes |b\rangle_2) = {}_1 \langle a| S_x^{(1)} |b\rangle_1 {}_2 \langle a|b\rangle_2.$$

Note that similarly to before, we must have $|a\rangle_2 = |b\rangle_2$, however also note that

$$S_x|+\rangle = \frac{\hbar}{2}|-\rangle, \quad S_x|-\rangle = \frac{\hbar}{2}|+\rangle.$$

We must thus have $|a\rangle_1 \neq |b\rangle_1$ (since the only options are $|+\rangle$ and $|-\rangle$). We thus have

$${}_1 \langle a| S_x^{(1)} |b\rangle_1 {}_2 \langle a|b\rangle_2 = \begin{cases} \hbar/2, & |a\rangle_1 \neq |b\rangle_1 \text{ and } |a\rangle_2 = |b\rangle_2 \\ 0, & \text{otherwise.} \end{cases}$$

A bit of playing with numbers yields that the matrix representing the $S_x^{(1)}$ term is given by

$$\frac{\hbar}{2} \begin{pmatrix} & & 1 & \\ & & & 1 \\ 1 & & & \\ & 1 & & \end{pmatrix}.$$

Similar logic shows that the matrix representing the $S_x^{(2)}$ term is given by

$$\frac{\hbar}{2} \begin{pmatrix} & 1 & \\ 1 & & \\ & & 1 \end{pmatrix}.$$

Putting everything together, we have

$$\begin{aligned} H &= -\frac{\hbar^2 J}{2} \begin{pmatrix} 1 & & \\ & -1 & \\ & & -1 \\ & & & 1 \end{pmatrix} + \frac{\hbar g}{2} \left(\begin{pmatrix} & 1 & \\ 1 & & \\ & & 1 \end{pmatrix} + \begin{pmatrix} & & 1 \\ 1 & & \\ & 1 & \end{pmatrix} \right) \\ &= -\frac{\hbar^2 J}{2} \begin{pmatrix} 1 & & \\ & -1 & \\ & & -1 \\ & & & 1 \end{pmatrix} + \frac{\hbar g}{2} \begin{pmatrix} & 1 & 1 & \\ 1 & & & 1 \\ 1 & & & 1 \\ & 1 & 1 & \end{pmatrix}. \end{aligned}$$

We can verify that this is correct by noticing that we are representing our basis states using the Kronecker product, so we can represent the dual space in the same way and immediately compute that

$$\begin{aligned} H &= -2J \frac{\hbar^2}{4} \begin{pmatrix} S_z & \\ & -S_z \end{pmatrix} + \frac{\hbar g}{2} \begin{pmatrix} & \mathbb{1} \\ \mathbb{1} & \end{pmatrix} + \frac{\hbar g}{2} \begin{pmatrix} \sigma_x & \\ & \sigma_x \end{pmatrix} \\ &= -\frac{\hbar^2 J}{2} \begin{pmatrix} 1 & & \\ & -1 & \\ & & -1 \\ & & & 1 \end{pmatrix} + \frac{\hbar g}{2} \begin{pmatrix} & 1 & 1 & \\ 1 & & & 1 \\ 1 & & & 1 \\ & 1 & 1 & \end{pmatrix}. \end{aligned}$$

This is exactly what we obtained in our calculations.

2. (a) The characteristic polynomial factors into

$$(1-x)(1+x) \left(\sqrt{4(g')^2 + 1} - x \right) \left(\sqrt{4(g')^2 + 1} + x \right).$$

Since g' is real, $(g')^2 > 0$, and thus $\sqrt{4(g')^2 + 1} > 1$. Therefore, $-\sqrt{4(g')^2 + 1}$ is the smallest nonzero eigenvalue of H .

For notational purposes, let

$$\Omega := \frac{1 - \sqrt{4(g')^2 + 1}}{2g'}.$$

We have

$$H = - \begin{pmatrix} 1 & & \\ & -1 & \\ & & -1 \\ & & & 1 \end{pmatrix} + g' \begin{pmatrix} & 1 & 1 & \\ 1 & & & 1 \\ 1 & & & 1 \\ & 1 & 1 & \end{pmatrix} = \begin{pmatrix} -1 & g' & g' & 0 \\ g' & 1 & 0 & g' \\ g' & 0 & 1 & g' \\ 0 & g' & g' & -1 \end{pmatrix}.$$

We also have

$$|\psi\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + \Omega \left(\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right) = \begin{pmatrix} 1 \\ \Omega \\ \Omega \\ 1 \end{pmatrix}.$$

By explicit computation, we have

$$H|\psi\rangle = \begin{pmatrix} -1 & g' & g' & 0 \\ g' & 1 & 0 & g' \\ g' & 0 & 1 & g' \\ 0 & g' & g' & -1 \end{pmatrix} \begin{pmatrix} 1 \\ \Omega \\ \Omega \\ 1 \end{pmatrix} = \begin{pmatrix} -1 + 2g'\Omega \\ 2g' + \Omega \\ 2g' + \Omega \\ -1 + 2g'\Omega \end{pmatrix}.$$

Note that our eigenvalue is $\lambda = -\sqrt{4(g')^2 + 1}$, which satisfies

$$\Omega = \frac{1 + \lambda}{2g'} \implies \lambda = 2g'\Omega - 1.$$

This immediately checks the top and bottom entries of $H|\psi\rangle$, so it suffices to show $\lambda\Omega = 2g' + \Omega$. We have

$$\begin{aligned} \left(-\sqrt{4(g')^2 + 1}\right) \left(\frac{1 - \sqrt{4(g')^2 + 1}}{2g'}\right) &= \frac{4(g')^2 + 1 - \sqrt{4(g')^2 + 1}}{2g'} \\ &= 2g' + \frac{1 - \sqrt{4(g')^2 + 1}}{2g'} \\ &= 2g' + \Omega. \end{aligned}$$

Therefore, $H|\psi\rangle = \lambda|\psi\rangle$.

(b) By our knowledge of limits as $g' \rightarrow \infty$,

$$\lim_{g' \rightarrow \infty} \frac{1 - \sqrt{1 + 4(g')^2}}{2g'} = \lim_{g' \rightarrow \infty} \frac{1 - \sqrt{4(g')^2}}{2g'} = \lim_{g' \rightarrow \infty} \frac{1 - 2g'}{2g'} = \lim_{g' \rightarrow \infty} \frac{-2g'}{2g'} = -1.$$

It follows that in the limit, we have

$$\begin{aligned} |\psi\rangle &= (|+\rangle_1 \otimes |+\rangle_2 + |-\rangle_1 \otimes |-\rangle_2) - (|+\rangle_1 \otimes |-\rangle_2 + |-\rangle_1 \otimes |+\rangle_2) \\ &= |+\rangle_1 \otimes (|+\rangle_2 - |-\rangle_2) + |-\rangle_1 \otimes (|-\rangle_2 - |+\rangle_2) \\ &= |+\rangle_1 \otimes |-\rangle_{x;2} + |-\rangle_1 \otimes (-|-\rangle_{x;2}) \\ &= |+\rangle_1 \otimes |-\rangle_{x;2} - |-\rangle_1 \otimes |-\rangle_{x;2} \\ &= (|+\rangle_1 - |-\rangle_1) \otimes |-\rangle_{x;2} \\ &= |-\rangle_{x;2} \otimes |-\rangle_{x;2}. \end{aligned}$$

If we let $g' \rightarrow 0$, then

$$\begin{aligned} \frac{1 - \sqrt{1 + 4(g')^2}}{2g'} &= \frac{1 - \sqrt{1 + 4(g')^2}}{2g'} \cdot \frac{1 + \sqrt{1 + 4(g')^2}}{1 + \sqrt{1 + 4(g')^2}} \\ &= \frac{1 - 1 - 4(g')^2}{2g' + 2g'\sqrt{1 + 4(g')^2}} \\ &= \frac{2g'}{1 + \sqrt{1 + 4(g')^2}} \\ &\xrightarrow{g' \rightarrow 0} \frac{0}{1 + \sqrt{1}} \\ &= 0. \end{aligned}$$

Therefore, in the limit we find

$$|\psi\rangle = |+\rangle_1 \otimes |+\rangle_2 + |-\rangle_1 \otimes |-\rangle_2.$$